

Savita Pareek

Postdoctoral Associate

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EDUCATION	Ph.D., Statistics, Indian Institute of Technology Bombay, India 2018 - 2023 <i>Dissertation:</i> On Some Problems in Mixed-effect Models, CPI: 9.0/10 <i>Advisor:</i> Prof. Siuli Mukhopadhyay M.Sc., Mathematical Sciences - Statistics 2014 - 2016 Banasthali University, India, Percentage Marks: 86.72/100 B.Sc., Mathematics, Physics, and Computer Science 2011 - 2014 Banasthali University, India, Percentage Marks: 84.06/100
PROFESSIONAL EXPERIENCE	Postdoctoral Associate, Massachusetts Institute of Technology, USA Jan 2026 – Present MIT Sloan School of Management and Schwarzman College of Computing (SERC) <i>Mentors:</i> Prof. Rahul Mazumder, Prof. Suzanne Flynn • Developing Interpretable ML methods for linguistic data to diagnose Alzheimer's. • Learning lab director at the Social and Ethical Responsibilities of Computing (SERC) program. Postdoctoral Research Fellow, Auburn University, USA Aug 2024 – Dec 2025 Department of Mathematics and Statistics; <i>Mentor:</i> Prof. Roberto Molinari • Developed differentially private equivalence tests for two-sample means and proportions. • Lead NIH R21 proposal on Rashomon-based inference for mental health disorders. Research Officer & Manager, Reserve Bank of India (RBI), Mumbai Feb 2021 – Aug 2024 Department of Statistics and Information Management • Designed an IFSC foreign exchange information-flow system, issuing an official circular. • Led compilation of millions of transaction-level records for India's Balance of Payments statistics. Ph.D. Research Projects, Indian Institute of Technology Bombay, India Jan 2018 - Oct 2023 • Mixed-effects models for crossover trials with missing and skewed responses. • Model-based clustering of gene expression time series, capturing temporal heterogeneity.
TEACHING & MENTORSHIP	Curriculum Developer, MIT Sloan School of Management, USA • Statistical Inference and Linear Models (PhD first-year course) 2026 (ongoing) Designing a new doctoral course; developing syllabus, lecture notes, and exercises. Instructor, Auburn University, USA • Fundamentals of Statistical Inference I (STAT 5510/6610, 11 students) Fall 2025 New graduate-level course; prepared original lectures, assignments, and assessments. Teaching evaluation (out of 6.0): mean 5.8, median 6.0 (response rate: 11/11) • Introduction to Data Science (STAT 4000, 46 students) Spring 2025 Second departmental offering; refined lectures, projects, and assessments. Teaching evaluation (out of 6.0): mean 3.2 (response rate: 13/46) • Probability & Statistics II (STAT 3610, 21 students) Fall 2024 Teaching evaluation (out of 6.0): median 4.0 (response rate: 13/21) Teaching Assistant, Indian Institute of Technology Bombay, India • Introduction to Probability (SI 427) Fall 2020 Led weekly tutorials; graded assignments and exams. • Regression Analysis (SI 422) Spring 2019, 2020 Conducted R programming labs; assisted student projects; graded exams. Mentorship • Mentor, NISS Writing Workshop for Junior Researchers Jul 2026 • Undergraduate student, Honors Contract project, Auburn University Spring 2025 • Graduate student, Rashomon sets in multi-model inference, Auburn University 2024-2025 • Research intern, India's current account sustainability, RBI Spring 2021 • Summer interns, Power BI dashboard and services trade, RBI Summer 2022, 2023 • Tutored Science & Math to students from underserved communities 2018-19 Teaching Certifications

- Kaufman Teaching Certificate Program, MIT Fall 2026 (ongoing)
- AUSSI Teaching Enhancement Program, Auburn University 2025

JOURNAL PUBLICATIONS

- Pareek, S.**, Insolia, L., Molinari, R., and Guerrier, S. (2026). *Equivalence Testing Under Privacy Constraints*. Under review (Statistics in Medicine), [arXiv]
- Eady, S., Ulloa, I. N., **Pareek, S.**, and Zabala, M. (2026). *The Effects of Psychosocial Stress on Surface Electromyography Signals of the Lower Leg During Isometric Contractions*. Under review.
- Pareek, S.** and Ghosh, S. (2026). *Semiparametric Dynamic Copula Models for Portfolio Optimization*. Journal of Data Science. [copula]
- Pareek, S.**, Das, K., and Mukhopadhyay, S. (2025). *Likelihood-based inference for skewed responses in a crossover trial setup*. Communications in Statistics - Simulation and Computation. [cssc]
- Pareek, S.**, Das, K., and Mukhopadhyay, S. (2023). *Likelihood-based missing data analysis in crossover trials*. Brazilian Journal of Probability and Statistics. [bjps]
- Pareek, S.**, Khetan, M., Mukhopadhyay, S., and Das, K. (2021). *Clustering gene expression time series data embedded in a non-parametric setup*. JSR (Special Issue). [jsr]

POLICY & BOOK CHAPTERS

- Misra, A., **Pareek, S.**, and Saggar, S. (2024). *Capturing e-commerce transactions from the travel group of BPM6: An Indian experience*. In External statistics in a fragmented and uncertain world (Vol. 62). [BIS]
- Biswas, D., **Pareek, S.**, and Saggar, S. (2022). *Freight costs of India's trade*. RBI Bulletin. [RBI]

SELECTED WORK IN PROGRESS

- Pareek, S.**, Lust, B., Flynn, S., and Mazumder, R. *Nature of relative clause production in amnesic mild cognitive impairment (aMCI)*.
- NIH R01 Proposal, MPI: **Pareek, S.**, and Molinari, R.; Co-PI: Deshpande, G. *Personalizing Mental Disorder Diagnosis with Rashomon Inference*.

HONORS & AWARDS

Research Awards

- Early Career Scholarship, Symposium on Data Science and Statistics (SDSS) 2025
- National Science Foundation (NSF) Travel Award, IISA Conference 2024
- Young Scientist Award, Society of Statistics and Computer Applications (SSCA), India 2024
- CWS Prof. Suddhendu Biswas Student Travel Award 2023
- International Indian Statistical Association (IISA) Student Travel Award 2022
- Gold Medal for Best Student in Science, Banasthali University, India 2016

Professional Activity Awards

- Certificate of Excellence, RBI Induction Program, India 2023
- Institute Organizational Special Mention, for IITB Hostel General Secretary leadership 2021
- Institute Sports Color, IIT Bombay, for excellence in competitive sports 2021
- Best Office Bearer, IITB Hostel 11, for exemplary service to the hostel community 2020

Competitive Exams

- 3rd in RBI Officer Exam (Statistics): among 200k 2020
- 6th in GATE (Mathematics): among 6k 2017
- 24th in CSIR-NET (Mathematical Sciences): among 40k 2017

PRESENTATIONS

Invited Talks

- Seminar Speaker, Indian Institute of Technology Bombay, India Jan 2026
- CFE-CMStatistics Conference, London (virtual) Dec 2025
- Seminar Speaker, Clemson University, USA (virtual) Sep 2025
- IISA International Conference on Statistics, Kochi, India Dec 2024
- External Statistics Conference, Banco de España, Madrid Feb 2024
- SSCA International Conference on Statistics, Banasthali, India Feb 2024
- Reserve Bank Staff College, Chennai, India (virtual) Oct 2023

Contributed Talks/Posters

- 10th Workshop on Biostatistics & Bioinformatics, Atlanta (poster) May 2025

- Symposium on Data Science & Statistics (SDSS), Salt Lake City May 2025
- ISPS International Conference on Statistics, Allahabad, India Feb 2024
- European Meeting of Statisticians (EMS), Poland (poster) Jul 2023
- ENAR Spring Meeting, Nashville, Tennessee Mar 2023
- IISA International Conference on Statistics, Bangalore, India Dec 2022

PROFESSIONAL ACTIVITIES

Journal Peer Review

- Journal of Statistical Theory and Practice, Journal of Investigative Medicine, Sankhya B

Service

- Session Chair: Topics in High-Dimensional and Network Data, IISA Conference 2024
- Contributed to ‘State of the Economy’ articles, RBI Monthly Bulletin 2022–2023
- Committee Member, Placement Cell, IIT Bombay 2021
- Executive Member, Post Graduate Academic Council, IIT Bombay 2020
- Volunteer for IWM and IISA conference, IIT Bombay 2019
- General Secretary, Hostel 11, IIT Bombay 2019
- Led yoga and wellness workshops, inspiring over 2k to live healthier 2020 - present

Membership

- American Statistical Association (ASA)
- Caucus for Women in Statistics & Data Science (CWS)
- International Indian Statistical Association (IISA)
- Society of Statistics, Computer & Applications (SSCA)

Training & Development

- Statistical Learning with Python, Stanford Online, edX 2025
- Preparing Future Faculty, Auburn University 2025
- Course (Re)Design for STEM Teaching, Auburn University 2024
- Compilation of National Accounts, 1-week, NSSTA, Noida, India 2022
- JAIIB Certification, Indian Institute of Banking & Finance 2022
- Remittance Statistics, 2-day, IMF’s Statistics Department (virtual) 2022
- Trade in Services Statistics, 6-week, UNSD, WTO, and UNCTAD (virtual) 2021
- BOP & IIP Statistics, 3-month, IMF Institute (virtual) 2021
- Power BI Data Analytics, 2-day, RBSC, Chennai, India 2021
- Yoga & Wellness Certifications (500-hr YTT; 1000-hr SKY Coach) 2021, 2025
- Mathematics for Data Science, 2-week, IISC Bangalore, India 2019

TECHNICAL SKILLS

- R, Python, Mathematica, and C++, $\text{\LaTeX}$